

SPR Optimizer: Comprehensive Project Report

This document provides a detailed technical analysis of the Surface Plasmon Resonance (SPR) Optimizer project. It covers the theoretical framework, computational methodology, optimization results, and a critical review of the codebase.

1. Project Description

The SPR Optimizer is a computational tool designed to automate the design of high-performance optical biosensors. The primary objective is to maximize the **Figure of Merit (FOM)**, a metric that balances high sensitivity (a large shift in resonance angle per unit of refractive index change) with a narrow resonance linewidth (Full Width at Half Maximum).

The system replaces traditional trial-and-error engineering with **Bayesian Optimization**. By simulating the physical interactions of light with multilayer structures, the software iteratively learns which material combinations and layer thicknesses yield the optimal sensor response.

2. Methodology & Physics

The Physics Engine (Transfer Matrix Method):

The core simulation engine uses the Transfer Matrix Method (TMM) to solve Maxwell's equations for a planar multilayer stack. For every candidate design, the system calculates the Fresnel reflection coefficients at each interface and cascades them to determine the total optical reflectance across a range of incident angles (typically 20° to 89°).

The Optimization Algorithm (Optuna):

The project utilizes the Optuna library to perform Bayesian hyperparameter optimization. The search space consists of four key parameters:

- **Metal Choice:** Selecting between Silver (Ag), Gold (Au), or Copper (Cu).
- **D2 (Metal Thickness):** Varied between 20 nm and 120 nm.
- **D3 (Dielectric Layer 1):** Varied between 0.5 nm and 20 nm.
- **D4 (Dielectric Layer 2):** Varied between 0.1 nm and 2 nm.

3. Results Analysis

Optimization Convergence:

Analysis of the optimization history reveals a rapid convergence profile. The algorithm identified a near-optimal solution (FOM ~187) within the first 50 trials. The remaining ~19,950 trials were spent exploring the parameter space but failed to find a superior configuration. This suggests the solution space has a distinct, steep global maximum that is easily discoverable.

Parameter Sensitivity:

A sensitivity analysis quantifying the impact of each variable on the final FOM score yielded the following hierarchy of importance:

- **1. Dielectric Layer 1 (D3) - 43% Importance:** The thickness of the PbMoO₄ layer is the single most critical factor. This layer likely acts to tune the propagation constant of the surface plasmon to match the prism coupling.
- **2. Metal Thickness (D2) - 30% Importance:** The thickness of the metal layer is crucial for determining the coupling efficiency and radiative damping of the plasmon.
- **3. Metal Choice - 13% Importance:** Surprisingly, the specific material (Ag vs Au) was less critical than the geometry of the layers.

Optimal Configuration Found:

The highest performing design identified by the system was a **Gold (Au)** based sensor. Specific parameters included a metal thickness of approximately **24.4 nm** and a primary dielectric layer of **13.5 nm**.

4. Technical Critique & Issues

While the results are promising, a code review identified three specific issues that may affect the validity of the findings:

- **Critical Bug (Edge Artifacts):** The FOM calculation uses ``argmin`` to find the resonance dip. If the dip shifts outside the simulated angular range, the algorithm selects the boundary index, resulting in a calculation of 'infinite' or artificially high sensitivity. This likely explains the unusually high FOM of 187 for Gold.
- **Objective Discontinuity:** The optimization loop switches between 'coarse' and 'fine' angular resolution based on a threshold score. This creates a step-function in the optimization landscape, which can confuse the Bayesian solver.
- **Numerical Instability:** The Fresnel equations used in the physics engine are susceptible to division-by-zero errors at the critical angle, which can lead to crashes or NaN values during simulation.